

Permissively Licensed Rooftop Array Detection at 21 cm, with an Executable Gate and Per-Array Roof Geometry from Public Lidar

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Working draft, not peer reviewed September 2026

Abstract

Draft, not a submission. This is the detection and roof-geometry half of a paper split because it was three papers in fifteen pages. The material below is moved from that draft and has not been rewritten to stand alone: it still refers forward to a soiling model that now lives in the other paper, and the introduction, related work and discussion are not written. It is released in this state so that the claims the companion paper cites can be checked, not as a finished document. See `SPLIT_PLAN.md`.

The contribution here is a detection gate stated as an executable definition rather than a number, a mask stage whose failure mode is a prompt-box problem rather than a mask problem, an independent recall probe against county building permits the detector never saw, and a projection trap in lidar-derived tilt that biases the angle by 18.9% on web-Mercator point clouds.

1 Methods

1.1 Array detection

Imagery.

Santa Cruz County publishes 2025 aerial orthoimagery as a free map service. We tile a pilot area of interest over the Live Oak and Pleasure Point side of the city into 249 tiles covering 14.93 km² of *actually imaged* ground; the enclosing bounding box is 23.71 km², and the distinction matters for any recall denominator. True ground GSD is 0.204 m. Tiles are stored in EPSG:3857, where the nominal pixel size reads 0.256 m, inflated by $1/\cos(37^\circ)$. Each tile records its CRS and affine transform, so any pixel-space detection projects back to geographic coordinates unambiguously. The county service natively serves 6.3 cm and we under-requested by $3.3\times$ linear, so the ceiling on this method sits above what we report.

Labels and splits.

Arrays are annotated as polygons, because array extent feeds the area-to-dollar chain and a box overstates it on non-rectangular roofs. The dataset comprises 976 chips (680

train / 100 validation / 196 test) and 3,894 chip-level polygons, rebuilt from 244 of the 249 tiles. **Two ground-truth counts exist and are not interchangeable:** chip-level (validation 385, test 636) double-counts arrays that the ~ 80 -pixel chip overlap repeats and splits those the seams divide, whereas tile-level (validation 340, test 585) does not. The gate scores tile-level; a tile-level F_1 quoted against 636 is wrong by construction.

Model and inference path.

RF-DETR at 728-pixel input, single class. Production inference decomposes each full tile into a 2×2 grid of 640-pixel chips, runs one forward pass per chip, and merges across seams by non-maximum suppression at IoU 0.55.

The gate, stated as an executable definition.

Our shipping criterion is *whatever scripts/detect/eval_tile_f1.py prints*: tile-level detection F_1 at box-IoU ≥ 0.50 , micro-averaged over the split, measured through the production path above. Three protocol choices define it.

1. **Confidence is tuned on validation and frozen; test is the reported number.** Tuning confidence on test spends the held-out split. The cost of not doing so is measured rather than assumed: test's own best F_1 is 0.8431 against 0.8260 at the validation-frozen threshold (Figure 1).
2. **Matching is on boxes, not masks**, so the gate is invariant to the presence of the mask stage. SAM2 changes area accuracy, not what was detected; conflate the two and a regression cannot be traced to its stage.
3. **The 95% interval bootstraps over tiles, not objects.** Arrays within a tile are correlated, and resampling objects yields an interval that is too narrow.

Passing requires all three of: $F_1 \geq 0.75$; F_1 95% CI lower bound ≥ 0.70 ; and recall ≥ 0.70 . Recall carries its own condition because a missed array is an unrecoverable missed lead, while the permit join filters false positives downstream.

Mask stage and prompt-box sensitivity.

SAM2 is in the product path not because the detector needs it, since the gate is box-based, but because *area is the product*. Prompted with the unmodified detector box, SAM2 returns whole-roof masks at a high rate. We treat the prompt box as the design variable and sweep it (§2.2).

1.2 Independent recall probe

Recall measured against held-out labels overstates recall against the real install base, because labels and model see the same imagery. We therefore cross-check detections against Santa Cruz County building-permit records, which the detector never trained on. Matching is by parcel APN rather than geocoded point. The Census batch geocoder interpolates each address onto the street segment, so points land a median 22 m from the nearest detected array and zero of 126 point matches fall inside a polygon. The point method measures geocode quality, not detection.

1.3 Roof geometry from public lidar

We fit a plane to USGS 3DEP [3] lidar returns inside each detected polygon, recovering per-array tilt and downslope azimuth over plain HTTP against the Entwine octree. The headline 3DEP product is a bare-earth DEM and is useless here by construction, since roofs are removed from it; the point cloud is not.

A projection trap.

Entwine X/Y are EPSG:3857 metres while Z is true metres. At this latitude the Mercator scale factor is $1/\cos(36.97^\circ) = 1.2515$, so a plane fitted on raw coordinates reports tilt **18.9% too shallow, silently, on every roof, producing an entirely plausible distribution**. We deflate X/Y by $\cos(\text{lat})$ and pin the magnitude in a unit test, so the correction cannot be removed unnoticed. Anyone reproducing this on web-Mercator lidar will hit it.

Failed fits are reported as null.

Arrays without a usable fit keep a flagged flat default and are never median-filled—precisely the failure mode §?? is about.

2 Results

2.1 Detection

Table 1 reports the gate. The shipping checkpoint passes all three conditions. Figure 1 shows the full operating curve on both splits and the size of the tuning penalty.

Table 1: Tile-level detection gate, $\text{box-IoU} \geq 0.50$, micro-averaged over the test split through the production chip-grid path. Confidence c^* tuned on validation and frozen. Intervals bootstrap over tiles ($n_{\text{tiles}} = 49$, $n_{\text{gt}} = 585$).

	W2 (shipped)	W1 (anchor)
c^* (val-frozen)	0.50	0.40
Precision	0.8499	0.8533
Recall	0.8034	0.7556
F_1	0.8260	0.8015
95% CI	[0.798, 0.853]	[0.773, 0.830]
Test-tuned best F_1	0.8431	0.8044
Gate verdict	PASS	PASS

Paired comparison, not overlapping intervals.

Doubling the training set moved F_1 from 0.8015 to 0.8260. The two marginal intervals overlap heavily and each point estimate lies inside the other’s interval, so the naive comparison reads “within noise.” That is the wrong test: both models were scored on the same 49 tiles, so the comparison is paired. Bootstrapping the per-tile difference gives $\Delta F_1 = +0.0246$, 95% CI $[+0.0051, +0.0441]$, $P(W2 > W1) = 0.993$, concentrated in recall $(+0.0479, \text{CI } [+0.0202, +0.0753], P = 0.999)$. The marginal comparison would have discarded a real gain.

Two caveats on the shipped checkpoint.

The tuning penalty grew relative to the anchor (0.017 against 0.003), so validation and test disagree more about where the operating point sits; and the chip-to-tile gap widened (0.029 against 0.008). Both are tracked and neither is yet explained.

Independent recall.

Against 434 geocoded permit points inside the imaged footprint, APN-matched recall is **55.3%** (240/434) across all years and **73.6%** (156/212) restricted to pre-cutoff installs that the imagery could contain. The point method reads 29.0% (Wilson CI 25.0–33.5%) for the geocoding reason given in §1.2. Three caveats travel with the number. Part of the gain could in principle come from spurious detections, since APN recall counts a parcel as found if any detected polygon’s interior point falls inside it; gate precision of 0.850 bounds but does not eliminate this. Permits are a valid recall probe and an *invalid* precision denominator: 85% of hand-labelled arrays sit on parcels with no permit on file, because the registry begins in 2016 and the residential boom preceded it. And 1,310 of 1,865 sites fall in city APN books with 0.0% county-permit coverage, because the City of Santa Cruz is a separate permitting authority. Any study joining detections to permits must check jurisdiction before reporting coverage.

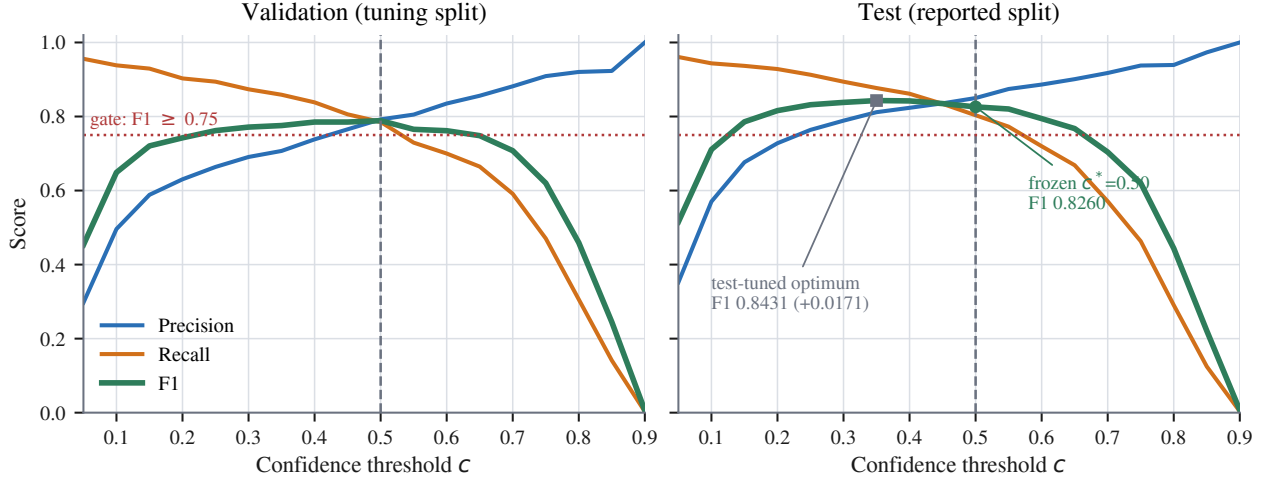


Figure 1: Operating curves on both splits. The frozen threshold $c^* = 0.50$ was chosen on validation alone; the test-tuned optimum sits 0.017 F_1 higher, which is the measured price of not spending the held-out split.

2.2 Mask quality is a prompt-box problem

Figure 2 sweeps the prompt box over 60 validation arrays. Prompting with the raw detector box yields median IoU 0.509, median mask area $1.97\times$ ground truth, and a 43.3% rate of whole-roof grabs. Shrinking the box 15% yields median IoU **0.844**, median area ratio **1.01**, and **0.0%** roof-grab. Area is therefore essentially unbiased, which is exactly what the area-to-kW-to-dollar chain needs.

Two principled alternatives, both measured, both rejected.

Background negative-point prompts behave exactly as theory predicts: they help when the box is bad (roof-grab $38.3\%\rightarrow 26.7\%$ at dilate-50%) and hurt when it is good ($0.844\rightarrow 0.820$, introducing 1.7% roof-grab where there was none), because with an already-shrunk box the ring sits near the true panel edge and clips real pixels. Multimask candidates clipped to the detector box with area-fit selection were worse everywhere ($0.824\rightarrow 0.673$ on exact boxes; roof-grab $1.7\%\rightarrow 71.7\%$ at dilate-25%), because picking the candidate whose area matches the detector box amplifies box error in exactly the regime the change targeted.

A ceiling we cannot locate.

We never measured inter-annotator agreement, because our annotation platform deduplicates and one tile cannot go to two labellers. If two humans agree to only ~ 0.85 IoU on a 26-pixel array, then 0.844 is the label-noise floor and further mask work is unmeasurable. We do not know which side of that line we are on.

2.3 Roof geometry

Plane fits succeed for 3,068 of 3,362 arrays (91.3%), with median RMS residual 0.040 m over ~ 270 points per array;

synthetic planes are recovered to $< 0.2^\circ$ tilt and $< 1^\circ$ azimuth. Figure 3 compares the recovered distribution against installer-reported tilt from 8,573 utility interconnection filings for the same county—a different instrument, a different decade, and no shared failure mode. **Both medians are 19.0° .** The low tails differ for a documented reason: the filings encode “not reported” as 0° and we drop those rows, so the filings understate the shallow tail (0.78% below 5°) while the lidar tail stays unbiased (8.38%). That 8.38% is directly relevant to Mejía and Kleissl [2]’s finding on sub-five-degree arrays.

What this comparison can and cannot establish.

It is distributional, not paired, and that is a property of the source rather than a choice: the DGStats interconnection extract carries a service city and nothing finer—no parcel number, address or coordinate—so no lidar array can be matched to the filing that describes it. 3D-PV-Locator [1] validates per system and is the stronger design where such a key exists.

We also decline to read agreement from the medians alone. A two-sample Kolmogorov–Smirnov test rejects decisively ($D = 0.214$, $p < 10^{-90}$, $n = 3,068$ against 8,573): the lidar distribution has much heavier tails, with a 10th percentile of 6.4° against 14° reported and a 90th of 30.8° against 27° . The mechanism is visible in the reported values. Every one is an integer, five values account for 57.6% of filings, and **27% are exactly 18°** —the 4:12 roof pitch. Installers file a nominal pitch from a small menu; the lidar fit measures a continuous angle on whatever roof is actually there. That the two medians land 0.03° apart across such different generating processes is the real result, and it is a statement about central tendency only. Per-array evidence, where we have it, comes instead from the standoff natural experiment below.

A natural experiment supports the interpretation: the li-

SAM2 prompt-box sensitivity (n=60 validation arrays, orange = production default)

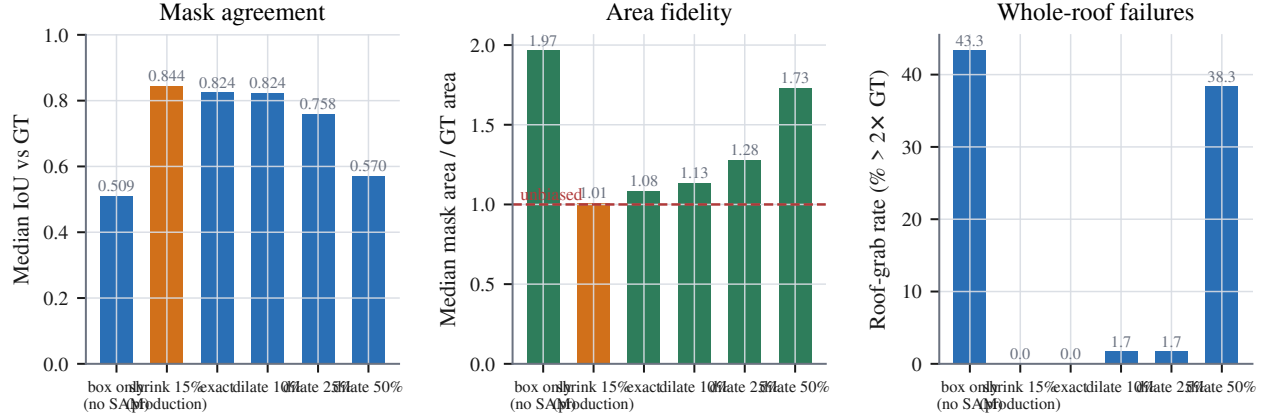


Figure 2: SAM2 prompt-box sensitivity. The whole-roof failure mode is created by an oversized prompt, not by the mask model, and disappears at the production default.

dar flight predates some installations and postdates others, and permit dates split the population on a variable unrelated to lidar. The panel standoff step appears exactly where physics says it should: pre-flight median +0.145 m against post-flight +0.013 m, difference +0.132 m, bootstrap 95% CI [+0.084, +0.208], $p < 0.0001$. The fit therefore reports true racking angle where the array predates the flight and roof pitch where it does not. The groups are small ($n = 37$ and 54) and only 15.9% of arrays yield a usable standoff.

References

- [1] Kevin Mayer, Benjamin Rausch, Marie-Louise Arlt, Gunther Gust, Zhecheng Wang, Dirk Neumann, and Ram Rajagopal. 3D-PV-Locator: Large-scale detection of rooftop-mounted photovoltaic systems in 3D. *Applied Energy*, 310:118469, 2022. doi: 10.1016/j.apenergy.2021.118469.
- [2] Felipe A. Mejía and Jan Kleissl. Soiling losses for solar photovoltaic systems in California. *Solar Energy*, 95: 357–363, 2013. doi: 10.1016/j.solener.2013.06.028.
- [3] U.S. Geological Survey. 3D elevation program (3DEP) lidar point cloud, 2020. Entwine Point Tile service. <https://www.usgs.gov/3d-elevation-program>.

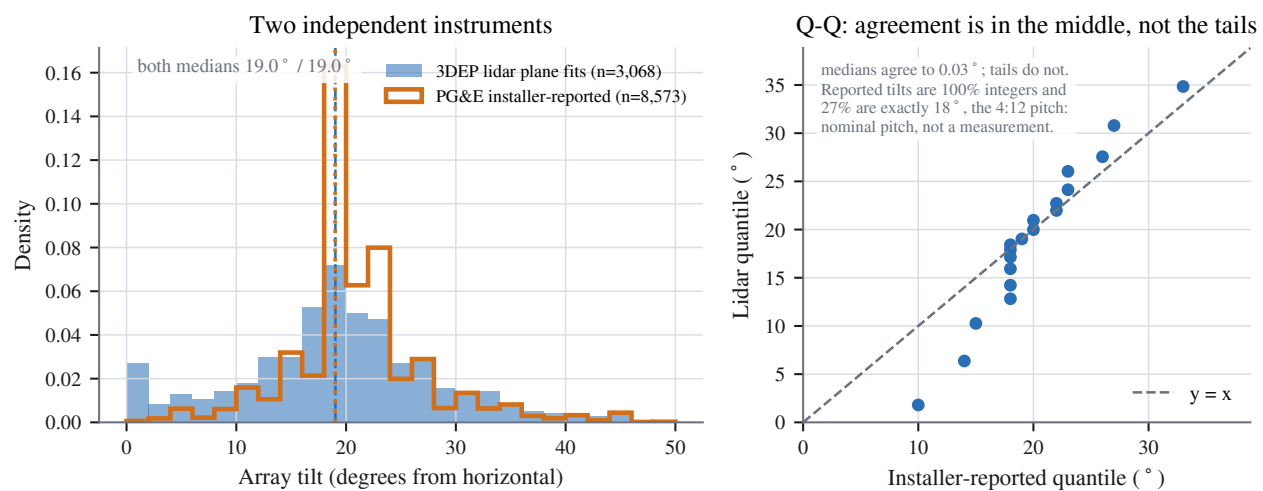


Figure 3: Per-array tilt from public lidar against installer-reported tilt from utility interconnection filings. The medians agree to 0.03° ; the distributions do not, and the reason is reporting granularity rather than measurement error.